

Alpilla 600 MW Combined Cycle Power Plant

KKS identification manual (project key list and tagging rules)

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Basis	requirement:ER-01.06

1. Purpose, scope and basis

This manual sets the plant identification of the project: systems, equipment, components, structures and the KKS part of every document number. It applies to the consortium members (Istanbul EPC, Imaginary Electric), their subsuppliers and subcontractors, for engineering, the 3D model, the engineering database, procurement, construction (AWP), commissioning and handover.

ER-01.06: All systems, equipment, instruments, cables and documents shall be identified according to KKS (VGB-B 105/106). The tag data shall be classified according to the AVEVA class library adopted by the Owner and delivered through the Owner's engineering database.

The identification follows VGB-S-811 (KKS) with the project key list of section 4. Agreement:

- TQ-035 (closed, answered 2026-09-28 by Owner): (1) Accepted: the consortium numbering is the project numbering under ER-17.01; the Owner's EDMS will use it unchanged. (2) The KKS key list is approved for engineering. The Owner's Engineer verifies each key title against the VGB-S-811 key catalogue with the KKS manual review; until then keys stay flagged 'not verified'. (3) MDL accepted as baseline for the Basic Design Package. Review classes accepted with one change: logic diagrams (LOG) incl. cause and effect are safety relevant and are for approval. The MDL is updated monthly with the progress report. (4) AWP definitions accepted. The Owner requires every CWP to be released for construction only when its EWP is complete (all IFC) and its PWP materials are on site or have a confirmed delivery before the planned installation.
- TQ-IEC-023 (closed, answered 2026-09-28 by Imaginary Electric): (1) Confirmed. IEC tags the power island with the project KKS to component level (B), systems and equipment units per the project key list; IEC part numbers are kept as vendor attributes below component level. IEC checks the key titles of its main groups against its VGB-S-811 licence copy and reports deviations with the KKS manual review (IFR). (2) Confirmed: project number primary, IEC number as vendor reference on the title block. (3) Confirmed: the VDRL dates in the MDL are the IEC commitment. (4) Confirmed with an improvement: generator shop tests are brought forward; the GT generators leave the works in week 49 together with the GT (sea transport 4 weeks), on site 2027-11-08. The ST generator follows the ST train (no change).

Decision: KKS per VGB-S-811 with the project key list (unit 0 common, 1 GT+HRSG, 2 ST); document numbers ALP-<ORG>-<KKS system>00000-<DISC>-<TYPE>-<NNNN>; IEC 61355 class per document type; supplier numbers as vendor reference only. (DEC-EPCE-0005, agreed, OWNER / IEPC / IEC)

2. Code structure

Codes are written without separators in the database and on documents (e.g. 10MBA10AA001). Data character types: A = letter, N = digit.

Level	Breakdown	Format	Meaning	Example
0	G	N	Plant unit (project numbering, table below)	1
1	F0	N	System prefix number (0 on this project)	0
1	F1 F2 F3	AAA	Function / system key (key list, section 4.1); U = structures	MBA
1	FN FN	NN	System number: 10, 20, ... parallel trains; units 11, 12 ... subdivisions	10
2	A1 A2	AA	Equipment unit key (section 4.2)	AP
2	AN AN AN	NNN	Equipment unit number: 001, 002 ... in flow direction	001
3	B1 B2	AA	Component key (section 4.3)	KP
3	BN BN	NN	Component number	01

Unit (G)	Plant unit
0	common / balance of plant
1	GT + HRSG island

Unit (G)	Plant unit
2	ST island

A system record is identified by G F0 F1F2F3 (e.g. 10MBA) or its group G F0 F1F2 (e.g. 10MB); the system number FN is part of equipment tags only. Examples from the plant asset register:

Tag	Breakdown
00EGB10BB001	unit 0 (common / balance of plant); system 00EGB10: Liquid fuel storage; equipment unit BB001: Storage equipment (vessels, tanks) - LDO storage tank A 3,500 m3
00EGB10BB002	unit 0 (common / balance of plant); system 00EGB10: Liquid fuel storage; equipment unit BB002: Storage equipment (vessels, tanks) - LDO storage tank B 3,500 m3
00GHA10AP001	unit 0 (common / balance of plant); system 00GHA10: Demineralised water storage and distribution; equipment unit AP001: Pump units - Demin water forwarding pump A
00ACA10	unit 0 (common / balance of plant); system 00ACA10: 380 kV switchgear (GIS) - 380 kV GIS, double busbar, 6 bays
00BBA10	unit 0 (common / balance of plant); system 00BBA10: MV switchgear 10.5 kV, normal supply - MV switchgear 10.5 kV, section A
00GHA10AP001KP01	unit 0 (common / balance of plant); system 00GHA10: Demineralised water storage and distribution; equipment unit AP001: Pump units; component KP01: Pump (hydraulic part) - component level (illustration)

3. Tagging rules

- Every system, item of equipment, instrument, electrical consumer, cable and structure receives a KKS code before it appears on a document; the code is created in the database first (engine CLI) and never typed on a drawing without a record.
- Unit 0 is used for common / balance-of-plant systems, 1 for the GT + HRSG island, 2 for the ST island; the unit of a system follows the train it serves, not its location.
- System numbers FN: 10 for the first train of a system, 20, 30 ... for parallel trains; 11, 12 ... for subdivisions of train 10. Equipment unit numbers AN: 001, 002 ... in flow direction; parallel units in the same system are numbered consecutively (e.g. pumps AP001 / AP002 for 2 x 100 %).
- Components (level B, e.g. motors KP / MA) carry the tag of their equipment unit plus B1B2 BN BN.
- Structures and buildings use the U function keys (section 4.1) with unit and system number, e.g. 10UMA.
- Measuring circuits and electrical equipment units use the A-keys of section 4.2; instrument tags follow the same level-2 structure (e.g. 10LAB10CP001).
- Supplier-internal part numbers are kept as vendor attributes below component level and never replace the KKS code (TQ-IEC-023).
- A code is never reused for a different item; deleted items keep their code void in the change log.
- Every tagged record carries its AVEVA class (ER-01.06) so the tag data can be exported to the Owner's engineering database (engine aveva_export).

4. Project key list

125 keys. 'Verified' means the key title has been checked against the licensed VGB-S-811 key catalogue by the Owner's Engineer (TQ-035); the key list itself is agreed for use.

4.1 Function keys (F1F2F3)

Key	Main group	Title	Status	Verified	Agreed in
ACA	Grid and distribution systems	380 kV switchgear (GIS)	agreed	no	TQ-035
BAA	Power transmission and auxiliary power supply	Generator leads (isolated phase busduct)	agreed	no	TQ-035
BAC	Power transmission and auxiliary power supply	Generator circuit breaker	agreed	no	TQ-035
BAT	Power transmission and auxiliary power supply	Generator step-up transformer	agreed	no	TQ-035
BBA	Power transmission and auxiliary power supply	MV switchgear 10.5 kV, normal supply	agreed	no	TQ-035
BBT	Power transmission and auxiliary power supply	Unit auxiliary transformer	agreed	no	TQ-035
BFA	Power transmission and auxiliary power supply	LV main switchgear 0.4 kV, normal supply	agreed	no	TQ-035
BFT	Power transmission and auxiliary power supply	LV transformers	agreed	no	TQ-035
BLA	Power transmission and auxiliary power supply	Lighting and small power	agreed	no	TQ-035
BMA	Power transmission and auxiliary power supply	LV switchgear, emergency supply	agreed	no	TQ-035
BRA	Power transmission and auxiliary power supply	UPS (uninterruptible AC)	agreed	no	TQ-035
BTA	Power transmission and auxiliary power supply	Battery systems and DC distribution 220 V	agreed	no	TQ-035
BXA	Power transmission and auxiliary power supply	Earthing and lightning protection	agreed	no	TQ-035

Key	Main group	Title	Status	Verified	Agreed in
CJA	Instrumentation and control	Plant distributed control system	agreed	no	TQ-035
CJG	Instrumentation and control	Safety instrumented systems	agreed	no	TQ-035
CKA	Instrumentation and control	Plant information, historian, performance monitoring, operator training simulator	agreed	no	TQ-035
CQA	Instrumentation and control	Continuous emission monitoring	agreed	no	TQ-035
CYA	Instrumentation and control	Communication, CCTV, access control and security	agreed	no	TQ-035
EGA	Conventional fuel supply and residues disposal	Liquid fuel unloading	agreed	no	TQ-035
EGB	Conventional fuel supply and residues disposal	Liquid fuel storage	agreed	no	TQ-035
EGC	Conventional fuel supply and residues disposal	Liquid fuel forwarding and treatment	agreed	no	TQ-035
EKA	Conventional fuel supply and residues disposal	Gas receiving, filtration and fiscal metering	agreed	no	TQ-035
EKC	Conventional fuel supply and residues disposal	Fuel gas distribution to consumers	agreed	no	TQ-035
GBA	Water supply and disposal	Seawater reverse osmosis (desalination)	agreed	no	TQ-035
GCA	Water supply and disposal	Demineralisation	agreed	no	TQ-035
GHA	Water supply and disposal	Demineralised water storage and distribution	agreed	no	TQ-035
GKA	Water supply and disposal	Potable water supply	agreed	no	TQ-035
GMA	Water supply and disposal	Process drains and waste water collection	agreed	no	TQ-035
GNA	Water supply and disposal	Waste water treatment	agreed	no	TQ-035
GQA	Water supply and disposal	Storm water drainage	agreed	no	TQ-035
GUA	Water supply and disposal	Sanitary waste water treatment	agreed	no	TQ-035
HAA	Conventional heat generation	HRSR pressure system (package)	agreed	no	TQ-035
HAN	Conventional heat generation	HRSR blowdown and drains	agreed	no	TQ-035
HNA	Conventional heat generation	Flue gas ducting	agreed	no	TQ-035
HNE	Conventional heat generation	Stack	agreed	no	TQ-035
HSA	Conventional heat generation	Selective catalytic reduction (SCR)	agreed	no	TQ-035
HSB	Conventional heat generation	Ammonia storage and supply	agreed	no	TQ-035
LAB	Steam, water, gas cycles	Feedwater piping	agreed	no	TQ-035
LAC	Steam, water, gas cycles	Boiler feed pumps	agreed	no	TQ-035
LAE	Steam, water, gas cycles	HP attemperation spray water	agreed	no	TQ-035
LAF	Steam, water, gas cycles	Reheat attemperation spray water	agreed	no	TQ-035
LBA	Steam, water, gas cycles	Main steam piping	agreed	no	TQ-035
LBB	Steam, water, gas cycles	Hot reheat piping	agreed	no	TQ-035
LBC	Steam, water, gas cycles	Cold reheat piping	agreed	no	TQ-035
LBD	Steam, water, gas cycles	LP steam piping	agreed	no	TQ-035
LBG	Steam, water, gas cycles	Auxiliary steam	agreed	no	TQ-035
LCA	Steam, water, gas cycles	Main condensate piping	agreed	no	TQ-035
LCB	Steam, water, gas cycles	Condensate extraction pumps	agreed	no	TQ-035
LDA	Steam, water, gas cycles	Condensate polishing	agreed	no	TQ-035
MAA	Main machine sets	HP turbine	agreed	no	TQ-035
MAB	Main machine sets	IP/LP turbine	agreed	no	TQ-035
MAG	Main machine sets	Condenser	agreed	no	TQ-035
MAJ	Main machine sets	Condenser vacuum system	agreed	no	TQ-035
MAN	Main machine sets	Turbine bypass stations	agreed	no	TQ-035
MAV	Main machine sets	ST lube oil system	agreed	no	TQ-035
MAW	Main machine sets	Gland steam system	agreed	no	TQ-035
MAX	Main machine sets	ST control fluid system	agreed	no	TQ-035
MAY	Main machine sets	ST control and protection	agreed	no	TQ-035
MBA	Main machine sets	Gas turbine (compressor, combustor, turbine)	agreed	no	TQ-035
MBL	Main machine sets	GT inlet air system	agreed	no	TQ-035
MBN	Main machine sets	GT liquid fuel and water injection	agreed	no	TQ-035
MBP	Main machine sets	GT fuel gas system incl. performance heater	agreed	no	TQ-035
MBU	Main machine sets	GT compressor washing	agreed	no	TQ-035
MBV	Main machine sets	GT lube oil system	agreed	no	TQ-035
MBY	Main machine sets	GT control and protection (TCS)	agreed	no	TQ-035
MKA	Main machine sets	Generator	agreed	no	TQ-035
MKC	Main machine sets	Excitation and static frequency converter	agreed	no	TQ-035
MKG	Main machine sets	Generator gas system (H2/CO2)	agreed	no	TQ-035
PAA	Cooling water systems	CW intake, screens and outfall	agreed	no	TQ-035
PAB	Cooling water systems	CW piping	agreed	no	TQ-035
PAC	Cooling water systems	CW pumps	agreed	no	TQ-035
PAH	Cooling water systems	Condenser tube cleaning	agreed	no	TQ-035
PBA	Cooling water systems	Electro-chlorination	agreed	no	TQ-035
PGA	Cooling water systems	Closed cooling water	agreed	no	TQ-035
QCA	Auxiliary systems	Chemical dosing	agreed	no	TQ-035
QEA	Auxiliary systems	Compressed air (service and instrument)	agreed	no	TQ-035
QHA	Auxiliary systems	Auxiliary boiler	agreed	no	TQ-035

Key	Main group	Title	Status	Verified	Agreed in
QJA	Auxiliary systems	Nitrogen supply	agreed	no	TQ-035
QJB	Auxiliary systems	Hydrogen and CO2 supply	agreed	no	TQ-035
QUA	Auxiliary systems	Sampling and steam/water analysis	agreed	no	TQ-035
SAA	Ancillary systems	Heating, ventilation and air conditioning	agreed	no	TQ-035
SGA	Ancillary systems	Fire water system	agreed	no	TQ-035
SGC	Ancillary systems	Deluge, spray and foam systems	agreed	no	TQ-035
SGY	Ancillary systems	Fire detection and alarm	agreed	no	TQ-035
SMA	Ancillary systems	Cranes and hoists	agreed	no	TQ-035
UAC	Structures	GIS building	agreed	no	TQ-035
UBA	Structures	Electrical building	agreed	no	TQ-035
UBF	Structures	Transformer bays	agreed	no	TQ-035
UCA	Structures	Control building	agreed	no	TQ-035
UEG	Structures	Liquid fuel tank farm structures	agreed	no	TQ-035
UEK	Structures	Gas station structures	agreed	no	TQ-035
UGA	Structures	Water treatment building	agreed	no	TQ-035
UHA	Structures	HRSR foundations and structures	agreed	no	TQ-035
UHN	Structures	Stack foundation	agreed	no	TQ-035
UMA	Structures	Gas turbine hall	agreed	no	TQ-035
UMB	Structures	Steam turbine hall	agreed	no	TQ-035
UPA	Structures	CW intake and outfall structures (marine)	agreed	no	TQ-035
UPC	Structures	CW pump house	agreed	no	TQ-035
UQA	Structures	Utility building (compressors, auxiliary boiler)	agreed	no	TQ-035
USA	Structures	Workshop and warehouse	agreed	no	TQ-035
UYA	Structures	Administration building	agreed	no	TQ-035
UZA	Structures	Roads and paving	agreed	no	TQ-035
UZB	Structures	Fencing and gates	agreed	no	TQ-035
UZA	Structures	Site drainage and culverts	agreed	no	TQ-035
XJA	Heavy machinery (not main machine sets)	Emergency diesel generator	agreed	no	TQ-035

4.2 Equipment unit keys (A1A2)

Key	Main group	Title	Status	Verified	Agreed in
AA	equipment unit	Valves, dampers	agreed	no	TQ-035
AC	equipment unit	Heat exchangers	agreed	no	TQ-035
AN	equipment unit	Compressor units, fans	agreed	no	TQ-035
AP	equipment unit	Pump units	agreed	no	TQ-035
AT	equipment unit	Cleaning, drying, filtering units	agreed	no	TQ-035
BB	equipment unit	Storage equipment (vessels, tanks)	agreed	no	TQ-035
BN	equipment unit	Jet pumps, ejectors	agreed	no	TQ-035
BP	equipment unit	Flow restrictors, orifices	agreed	no	TQ-035
BR	equipment unit	Piping, ducting	agreed	no	TQ-035
BS	equipment unit	Silencers	agreed	no	TQ-035
CF	equipment unit	Direct measuring circuit - flow	agreed	no	TQ-035
CL	equipment unit	Direct measuring circuit - level	agreed	no	TQ-035
CP	equipment unit	Direct measuring circuit - pressure	agreed	no	TQ-035
CQ	equipment unit	Direct measuring circuit - analysis	agreed	no	TQ-035
CS	equipment unit	Direct measuring circuit - speed	agreed	no	TQ-035
CT	equipment unit	Direct measuring circuit - temperature	agreed	no	TQ-035
CY	equipment unit	Direct measuring circuit - vibration	agreed	no	TQ-035

4.3 Component keys (B1B2)

Key	Main group	Title	Status	Verified	Agreed in
KA	component	Valve body / gate and globe valve component	agreed	no	TQ-035
KP	component	Pump (hydraulic part)	agreed	no	TQ-035
MA	component	Electric motor	agreed	no	TQ-035

5. System register

116 systems and groups. Scope = party responsible for the design of the system.

Code	Title	Category	Scope	Parent	AVEVA class
00ACA	380 kV GIS switchyard	electrical hv	IEPC	-	System
00BBA	MV switchgear 10.5 kV	electrical mv lv	IEPC	-	System
00BFA	LV switchgear 0.4 kV	electrical mv lv	IEPC	-	System
00BFT	LV transformers	electrical mv lv	IEPC	-	System
00BLA	Lighting and small power	electrical mv lv	IEPC	-	System
00BMA	Emergency LV switchgear	electrical mv lv	IEPC	-	System
00BRA	UPS	electrical mv lv	IEPC	-	System
00BTA	Batteries and DC 220 V	electrical mv lv	IEPC	-	System
00BXA	Earthing and lightning protection	common	IEPC	-	System
00CJA	Plant DCS	i and c	IEPC	-	System
00CJG	Safety instrumented systems (BOP)	i and c	IEPC	-	System
00CKA	Historian, performance monitoring and operator training simulator	i and c	IEPC	-	System
00CQA	Continuous emission monitoring	i and c	IEPC	-	System
00CYA	Telecom, CCTV, access control, security	i and c	IEPC	-	System
00EGA	LDO unloading	fuel gas	IEPC	-	System
00EGB	LDO storage	fuel gas	IEPC	-	System

Code	Title	Category	Scope	Parent	AVEVA class
00EGC	LDO forwarding and treatment	fuel gas	IEPC	-	System
00EKA	Gas receiving, filtration and fiscal metering station	fuel gas	IEPC	-	System
00EKC	Fuel gas distribution to GT and auxiliary boiler	fuel gas	IEPC	-	System
00GBA	Seawater reverse osmosis	water treatment	IEPC	-	System
00GCA	Demineralisation	water treatment	IEPC	-	System
00GHA	Demineralised water storage and distribution	water treatment	IEPC	-	System
00GKA	Potable water	water treatment	IEPC	-	System
00GMA	Process drains and waste water collection	water treatment	IEPC	-	System
00GNA	Waste water treatment	water treatment	IEPC	-	System
00GQA	Storm water drainage	civil structural	IEPC	-	System
00GUA	Sanitary waste water treatment	water treatment	IEPC	-	System
00LBG	Auxiliary steam	water steam cycle	IEPC	-	System
00LDA	Condensate polishing	water treatment	IEPC	-	System
00PAA	CW intake, screens and outfall	cooling	IEPC	-	System
00PAB	CW piping	cooling	IEPC	-	System
00PAC	CW pumps	cooling	IEPC	-	System
00PAH	Condenser tube cleaning	cooling	IEPC	-	System
00PBA	Electro-chlorination	cooling	IEPC	-	System
00PGA	Closed cooling water	cooling	IEPC	-	System
00QCA	Chemical dosing	auxiliary mechanical	IEPC	-	System
00QEA	Compressed air	auxiliary mechanical	IEPC	-	System
00QHA	Auxiliary boiler	auxiliary mechanical	IEPC	-	System
00QJA	Nitrogen supply	auxiliary mechanical	IEPC	-	System
00QJB	Hydrogen and CO2 supply	auxiliary mechanical	IEPC	-	System
00QUA	Sampling and SWAS	auxiliary mechanical	IEPC	-	System
00SAA	HVAC	hvac	IEPC	-	System
00SGA	Fire water system	fire protection	IEPC	-	System
00SGC	Deluge, spray and foam systems	fire protection	IEPC	-	System
00SGY	Fire detection and alarm	i and c	IEPC	-	System
00SMA	Cranes and hoists	other	IEPC	-	System
00UAC	GIS building	civil structural	IEPC	-	System
00UBA	Electrical building	civil structural	IEPC	-	System
00UBF	Transformer bays	civil structural	IEPC	-	System
00UCA	Control building	civil structural	IEPC	-	System
00UEG	LDO tank farm structures	civil structural	IEPC	-	System
00UEK	Gas station structures	civil structural	IEPC	-	System
00UGA	Water treatment building and basins	civil structural	IEPC	-	System
00UPA	CW intake and outfall marine structures	civil structural	IEPC	-	System
00UPC	CW pump house	civil structural	IEPC	-	System
00UQA	Utility building	civil structural	IEPC	-	System
00USA	Workshop and warehouse	civil structural	IEPC	-	System
00UYA	Administration building	civil structural	IEPC	-	System
00UZA	Roads and paving	civil structural	IEPC	-	System
00UZB	Fencing and gates	civil structural	IEPC	-	System
00UZC	Site drainage and culverts	civil structural	IEPC	-	System
00XJA	Emergency diesel generator	electrical mv lv	IEPC	-	System
10BAA	GT generator busduct	electrical hv	IEC	-	System
10BAC	GT generator circuit breaker	electrical hv	IEC	-	System
10BAT	GT generator step-up transformer	electrical hv	-	-	System
10BBT	GT unit auxiliary transformer	electrical mv lv	IEPC	-	System
10HA	HRSG pressure system	hrsg	IEC	-	System
10HAA	HRSG pressure system	hrsg	IEC	10HA	System
10HAN	HRSG blowdown and drains (EPC tanks and headers)	water steam cycle	IEPC	10HA	System
10HNA	HRSG flue gas ducting	hrsg	IEC	10HA	System
10HNE	Stack 65 m with CEMS platform and silencer	hrsg	IEPC	-	System
10HSA	SCR catalyst and ammonia injection	hrsg	IEC	-	System
10HSB	Ammonia storage and supply	auxiliary mechanical	IEPC	-	System
10LAB	Feedwater piping	condensate feedwater	IEPC	-	System
10LAC	Boiler feed pumps	condensate feedwater	IEPC	-	System
10LAE	HP attemperation spray water	condensate feedwater	IEPC	-	System
10LAF	Reheat attemperation spray water	condensate feedwater	IEPC	-	System
10LBA	Main steam piping	water steam cycle	IEPC	-	System
10LBB	Hot reheat piping	water steam cycle	IEPC	-	System
10LBC	Cold reheat piping	water steam cycle	IEPC	-	System
10LBD	LP steam piping	water steam cycle	IEPC	-	System
10MB	Gas turbine plant	gas turbine	IEC	-	System
10MBA	Gas turbine	gas turbine	IEC	10MB	System
10MBL	GT inlet air system	gas turbine	IEC	10MB	System
10MBN	GT liquid fuel and water injection	gas turbine	IEC	10MB	System
10MBP	GT fuel gas system incl. performance heater	gas turbine	IEC	10MB	System
10MBU	GT compressor washing	gas turbine	IEC	10MB	System
10MBV	GT lube oil system	gas turbine	IEC	10MB	System
10MBY	GT control and protection (TCS)	i and c	IEC	10MB	System
10MK	GT generator plant	generator	IEC	-	System
10MKA	GT generator	generator	IEC	10MK	System
10MKC	GT excitation and static frequency converter	generator	IEC	10MK	System

Code	Title	Category	Scope	Parent	AVEVA class
10MKG	GT generator gas system (H2/CO2)	generator	IEC	10MK	System
10UHA	HRSG foundations and structures	civil structural	IEPC	-	System
10UHN	Stack foundation	civil structural	IEPC	-	System
10UMA	Gas turbine hall	civil structural	IEPC	-	System
20BAA	ST generator busduct	electrical hv	IEC	-	System
20BAC	ST generator circuit breaker	electrical hv	IEC	-	System
20BAT	ST generator step-up transformer	electrical hv	-	-	System
20BBT	ST unit auxiliary transformer	electrical mv lv	IEPC	-	System
20LCA	Main condensate piping	condensate feedwater	IEPC	-	System
20LCB	Condensate extraction pumps	condensate feedwater	IEPC	-	System
20MA	Steam turbine plant	steam turbine	IEC	-	System
20MAA	HP turbine	steam turbine	IEC	20MA	System
20MAB	IP/LP turbine	steam turbine	IEC	20MA	System
20MAG	Condensing system	water steam cycle	-	-	System
20MAJ	Condenser vacuum system	steam turbine	IEC	20MA	System
20MAN	Turbine bypass stations and piping	water steam cycle	IEPC	20MA	System
20MAV	ST lube oil system	steam turbine	IEC	20MA	System
20MAW	Gland steam system	steam turbine	IEC	20MA	System
20MAX	ST control fluid system	steam turbine	IEC	20MA	System
20MAY	ST control and protection	i and c	IEC	20MA	System
20MK	ST generator plant	generator	IEC	-	System
20MKA	ST generator	generator	IEC	20MK	System
20MKC	ST excitation	generator	IEC	20MK	System
20UMB	Steam turbine hall	civil structural	IEPC	-	System

6. Plant asset register (equipment tags)

76 equipment items. CWA / CWP / MR assign each asset to its construction area, construction package and procurement package (AWP, ALP-EPC-00000-GE-PLN-0002).

Tag	Description	AVEVA class	Supply	CWA	CWP	MR
00ACA10	380 kV GIS, double busbar, 6 bays	Switchgear And Controlgear	IEPC	CWA-04	CWP-04-EL-01	MR-EL-001
00BBA10	MV switchgear 10.5 kV, section A	Switchgear And Controlgear	IEPC	CWA-03	CWP-03-EL-01	MR-EL-003
00BBA20	MV switchgear 10.5 kV, section B	Switchgear And Controlgear	IEPC	CWA-03	CWP-03-EL-01	MR-EL-003
00BFA10	LV main switchboard 0.4 kV, A	Switchgear And Controlgear	IEPC	CWA-03	CWP-03-EL-01	MR-EL-005
00BFA20	LV main switchboard 0.4 kV, B	Switchgear And Controlgear	IEPC	CWA-03	CWP-03-EL-01	MR-EL-005
00BFT10	LV transformer 10.5/0.4 kV 2.5 MVA, A	Power Transformer	IEPC	CWA-03	CWP-03-EL-01	MR-EL-004
00BFT20	LV transformer 10.5/0.4 kV 2.5 MVA, B	Power Transformer	IEPC	CWA-03	CWP-03-EL-01	MR-EL-004
00BMA10	Emergency LV switchboard	Switchgear And Controlgear	IEPC	CWA-03	CWP-03-EL-01	MR-EL-005
00BRA10	UPS 230 V AC (2 x 100 %)	UPS Unit	IEPC	CWA-03	CWP-03-EL-01	MR-EL-007
00BTA10	Battery and charger 220 V DC (2 x 100 %)	Battery Back-Up Power Supply Unit	IEPC	CWA-03	CWP-03-EL-01	MR-EL-007
00CJA10	Plant DCS (controllers, networks, operator stations)	Control Cabinet	IEPC	CWA-11	CWP-11-IC-01	MR-IC-001
00CQA10	Continuous emission monitoring system	Analyser	IEPC	CWA-01	CWP-01-IC-01	MR-IC-002
00EGA10	LDO unloading station (4 bays)	Equipment Package Skid	IEPC	CWA-10	CWP-10-ME-01	MR-ME-016
00EGB10BB001	LDO storage tank A 3,500 m3	Fixed Roof Tank	IEPC	CWA-10	CWP-10-ME-01	MR-ME-017
00EGB10BB002	LDO storage tank B 3,500 m3	Fixed Roof Tank	IEPC	CWA-10	CWP-10-ME-01	MR-ME-017
00EGC10	LDO forwarding and treatment skid	Equipment Package Skid	IEPC	CWA-10	CWP-10-ME-01	MR-ME-016
00EKA10	Gas receiving, filtration and fiscal metering station	Equipment Package Skid	IEPC	CWA-09	CWP-09-ME-01	MR-ME-015
00GBA10	Seawater reverse osmosis package (2 x 100 %)	Equipment Package Skid	IEPC	CWA-07	CWP-07-ME-01	MR-ME-012
00GCA10	Demineralisation package (EDI / mixed bed)	Equipment Package Skid	IEPC	CWA-07	CWP-07-ME-01	MR-ME-012
00GHA10AP001	Demin water forwarding pump A	Centrifugal Pump	IEPC	CWA-07	CWP-07-ME-01	MR-ME-013
00GHA10AP002	Demin water forwarding pump B	Centrifugal Pump	IEPC	CWA-07	CWP-07-ME-01	MR-ME-013
00GHA10BB001	Demineralised water tank A	Tank	IEPC	CWA-07	CWP-07-ME-01	MR-ME-013
00GHA10BB002	Demineralised water tank B	Tank	IEPC	CWA-07	CWP-07-ME-01	MR-ME-013
00GKA10BB001	Potable water tank	Tank	IEPC	CWA-07	CWP-07-ME-01	MR-ME-013
00GNA10	Waste water treatment package	Equipment Package Skid	IEPC	CWA-07	CWP-07-ME-01	MR-ME-014
00GUA10	Sanitary treatment package	Equipment Package Skid	IEPC	CWA-07	CWP-07-ME-01	MR-ME-014
00LDA10	Condensate polishing plant	Equipment Package Skid	IEPC	CWA-07	CWP-07-ME-01	MR-ME-012
00PAA10AT001	Travelling band screen A	Filter	IEPC	CWA-05	CWP-05-ME-01	MR-ME-004
00PAA10AT002	Travelling band screen B	Filter	IEPC	CWA-05	CWP-05-ME-01	MR-ME-004
00PAC10AP001	CW pump A (2 x 50 %, vertical)	Centrifugal Pump	IEPC	CWA-05	CWP-05-ME-01	MR-ME-003
00PAC10AP002	CW pump B (2 x 50 %, vertical)	Centrifugal Pump	IEPC	CWA-05	CWP-05-ME-01	MR-ME-003
00PAH10	Condenser tube cleaning system (balls, strainers)	Equipment Package Skid	IEPC	CWA-02	CWP-02-ME-01	MR-ME-006

Tag	Description	AVEVA class	Supply	CWA	CWP	MR
00PBA10	Electro-chlorination package	Equipment Package Skid	IEPC	CWA-05	CWP-05-ME-01	MR-ME-005
00PGA10AC001	CCW plate heat exchanger A (2 x 100 %)	Plate Heat Exchanger	IEPC	CWA-08	CWP-08-ME-01	MR-ME-007
00PGA10AC002	CCW plate heat exchanger B (2 x 100 %)	Plate Heat Exchanger	IEPC	CWA-08	CWP-08-ME-01	MR-ME-007
00PGA10AP001	CCW pump A (2 x 100 %)	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-007
00PGA10AP002	CCW pump B (2 x 100 %)	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-007
00PGA10BB001	CCW expansion tank	Pressure Vessel	IEPC	CWA-08	CWP-08-ME-01	MR-ME-007
00QCA10	Chemical dosing skid (ammonia, oxygen scavenger)	Equipment Package Skid	IEPC	CWA-08	CWP-08-ME-01	MR-ME-011
00QEA10AN001	Air compressor A (2 x 100 %)	Compressor	IEPC	CWA-08	CWP-08-ME-01	MR-ME-008
00QEA10AN002	Air compressor B (2 x 100 %)	Compressor	IEPC	CWA-08	CWP-08-ME-01	MR-ME-008
00QEA10AT001	Instrument air dryer (2 x 100 %)	Instrument Air Dryer	IEPC	CWA-08	CWP-08-ME-01	MR-ME-008
00QEA10BB001	Instrument air receiver	Pressure Vessel	IEPC	CWA-08	CWP-08-ME-01	MR-ME-008
00QHA10	Auxiliary boiler 12 t/h, gas-fired	Boiler	IEPC	CWA-08	CWP-08-ME-01	MR-ME-009
00QJA10	Nitrogen supply package (bottle racks / generator)	Equipment Package Skid	IEPC	CWA-08	CWP-08-ME-01	MR-ME-010
00QJB10	Hydrogen and CO2 bottle racks	Equipment Package Skid	IEPC	CWA-01	CWP-01-ME-01	MR-ME-010
00QUA10	Steam/water analysis system (SWAS)	Equipment Package Skid	IEPC	CWA-01	CWP-01-ME-01	MR-IC-003
00SAA10	HVAC plant (chillers, AHUs, pressurisation)	Equipment Package Skid	IEPC	CWA-03	CWP-03-HV-01	MR-HV-001
00SGA10AP001	Electric fire pump	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-020
00SGA10AP002	Diesel fire pump	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-020
00SGA10AP003	Jockey pump	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-020
00SGA10BB001	Fire water tank	Fixed Roof Tank	IEPC	CWA-08	CWP-08-ME-01	MR-ME-013
00SMA10	GT hall EOT crane 120/20 t	Crane	IEPC	CWA-01	CWP-01-SS-01	MR-ME-021
00SMA20	ST hall EOT crane 100 t	Crane	IEPC	CWA-02	CWP-02-SS-01	MR-ME-021
00XJA10	Emergency diesel generator 1,250 kVA	Synchronous Generator	IEPC	CWA-03	CWP-03-EL-01	MR-EL-006
10BAT10	GT step-up transformer 560 MVA 400/21 kV	Power Transformer	IEC	CWA-01	CWP-01-EL-01	MR-EL-102
10BBT10	GT unit auxiliary transformer 21/10.5 kV 31.5 MVA	Power Transformer	IEPC	CWA-01	CWP-01-EL-01	MR-EL-002
10HAA10	Heat recovery steam generator, 3-pressure reheat	Steam Generator	IEC	CWA-01	CWP-01-ME-01	MR-ME-102
10HAN10BB001	HRSB blowdown tank	Tank	IEPC	CWA-01	CWP-01-ME-01	MR-ME-019
10HAN10BB002	Atmospheric drain tank	Tank	IEPC	CWA-01	CWP-01-ME-01	MR-ME-019
10HNE10	Stack 65 m, D 9.2 m, with silencer and CEMS platform	Stack	IEPC	CWA-01	CWP-01-ME-01	MR-SS-002
10HSB10AP001	Ammonia forwarding pump A	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-018
10HSB10AP002	Ammonia forwarding pump B	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-018
10HSB10BB001	Aqueous ammonia tank A 30 m3	Tank	IEPC	CWA-08	CWP-08-ME-01	MR-ME-018
10HSB10BB002	Aqueous ammonia tank B 30 m3	Tank	IEPC	CWA-08	CWP-08-ME-01	MR-ME-018
10LAC10AP001	Boiler feed pump A (2 x 100 %, VFD)	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-001
10LAC10AP002	Boiler feed pump B (2 x 100 %, VFD)	Centrifugal Pump	IEPC	CWA-08	CWP-08-ME-01	MR-ME-001
10MBA10	Gas turbine	Gas Turbine	IEC	CWA-01	CWP-01-ME-01	MR-ME-101
10MKA10	GT generator 560 MVA, H2-cooled	Synchronous Generator	IEC	CWA-01	CWP-01-ME-01	MR-EL-101
20BAT10	ST step-up transformer 270 MVA 400/15.75 kV	Power Transformer	IEC	CWA-02	CWP-02-EL-01	MR-EL-102
20BBT10	ST unit auxiliary transformer 15.75/10.5 kV 16 MVA	Power Transformer	IEPC	CWA-02	CWP-02-EL-01	MR-EL-002
20LCB10AP001	Condensate extraction pump A (2 x 100 %)	Centrifugal Pump	IEPC	CWA-02	CWP-02-ME-01	MR-ME-002
20LCB10AP002	Condensate extraction pump B (2 x 100 %)	Centrifugal Pump	IEPC	CWA-02	CWP-02-ME-01	MR-ME-002
20MAA10	Steam turbine, reheat condensing, 2 casings	Steam Turbine	IEC	CWA-02	CWP-02-ME-01	MR-ME-103
20MAG10	Surface condenser, titanium tubes, 15,200 m2	Condenser	IEC	CWA-02	CWP-02-ME-01	MR-ME-103
20MKA10	ST generator 270 MVA, TEWAC	Synchronous Generator	IEC	CWA-02	CWP-02-ME-01	MR-EL-101

7. Tools and control

- Check or explain a code: `python -m engine kks <tag | system | document number>`.
- Create tags only via `python -m engine db add equipment ... --reason (classify with --set aveva_class=...)`.
- The `engineering_plan` engine checks every system code and equipment tag against the key list at each run; findings are listed below and in the MDL workbook.
- Changes to the key list: new key record (status proposed) -> Owner agreement by TQ -> status agreed.

Check results (engine)

3 OK, 1 INFO, 0 WARN (checks: AVEVA class, KKS key list, KKS system, KKS tag). WARN must be resolved before the next issue; INFO is noted.

Item	Check	Result	Detail
-	KKS key list	INFO	125 key titles not yet verified against the licensed VGB key list
-	AVEVA class	OK	76 of 76 equipment records classified
-	AVEVA class	OK	116 of 116 system records classified
-	AVEVA class	OK	623 of 623 document records classified

